

Introduction to Snowflake SQL

INTRODUCTION TO SNOWFLAKE SQL



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Connecting to Snowflake

- Snowsight: Snowflake Web Interface

The screenshot displays the Snowflake Snowsight web interface. On the left, a sidebar shows the 'Databases' tab selected, with a tree view of the database structure. The 'PIZZA_SALES' database is expanded, showing the 'PUBLIC' schema and a list of tables: 'ORDERS', 'ORDERS_FILTERED', 'ORDER_DETAILS', 'PIZZAS', and 'PIZZA_TYPE'. The 'ORDERS_VIEW' is also visible under the 'Views' section. The main area shows a worksheet named 'test_worksheet' with a SQL query: `select :datebucket(created), count(1) from table group by 1`. Below the query, the 'Results' tab is active, displaying a table with 5 columns: 'C_CUSTOMER_SK', 'C_CUSTOMER_ID', 'C_CURRENT_CDEMO_SK', 'C_CURRENT_', and 'C_CURRENT_'. The table contains 4 rows of data.

	C_CUSTOMER_SK	C_CUSTOMER_ID	C_CURRENT_CDEMO_SK	C_CURRENT_	C_CURRENT_
1	68,459,673	AAAAAAAAAJJMJEBA	1,323,178		
2	68,459,674	AAAAAAAKJMJEBEA	1,828,933		
3	68,459,675	AAAAAAAALJMJEBEA	404,253		
4	68,459,676	AAAAAAAAMJMJEBEA	145,624		

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Notebooks

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notebook_app.ipynb

...

Python ▾ as cell1 •

1

Import python packages

2

import pandas as pd

3

4

We can also use Snowpark for our analyses!

5

from snowflake.snowpark.context import import get_active_session

6

session = get_active_session()

7

• cell1

cell2

⋮

SQL ▾ as cell2

1.3s ▶ ☰ ⋮

1

-- Welcome to Snowflake Notebooks!

2

-- Try out a SQL cell to generate some data.

3

SELECT *

4

FROM course_read_33509_prod.pizza.pizzas

	PIZZA_ID	PIZZA_TYPE_ID	PIZZA_SIZE	PRICE
0	bbq_ckn_s	bbq_ckn	S	12.75
1	bbq_ckn_m	bbq_ckn	M	16.75
2	bbq_ckn_l	bbq_ckn	L	20.75
3	cali_ckn_s	cali_ckn	S	12.75
4	cali_ckn_m	cali_ckn	M	16.75
5	cali_ckn_l	cali_ckn	L	20.75

Connecting to Snowflake: Drivers

Drivers & Connectors

- ODBC (Open Database Connectivity)
- JDBC (Java Database Connectivity)
- Connectors : Python, Spark, and more

¹ <https://docs.snowflake.com/en/developer-guide/drivers>

Connecting to Snowflake: SnowSQL

SnowSQL

- Command-line client
 - Installed on Linux, Windows, or Mac

¹ <https://docs.snowflake.com/en/user-guide/snowsql>

SQL flavors

- Snowflake uses Snowflake SQL
- Other popular SQL flavors:
 - PostgreSQL
 - T-SQL
 - MySQL
- Some differences in data types, functions, and general syntax

Common syntax

- `SELECT`
- `FROM`
- `WHERE`
- `GROUPBY`
- `ORDER BY`
- `AVG()` , `COUNT()` , `SUM()` (etc)
- `DISTINCT`
- `HAVING`
- `JOIN *`

*Some differences exist, more on this later!

Let's practice!

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Snowflake SQL data types

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Common data types

Category	Data types
Text/string	<code>VARCHAR</code> , <code>CHAR</code> , <code>TEXT</code>

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Common data types

Category	Data types
Text/string	<code>VARCHAR</code> , <code>CHAR</code> , <code>TEXT</code>
Numeric	<code>INTEGER</code>
Boolean	<code>BOOLEAN</code>
Date/time	<code>DATE</code> , <code>TIME</code> , <code>TIMESTAMP</code>

¹ <https://docs.snowflake.com/en/sql-reference/intro-summary-data-types>

Snowflake SQL data types - NUMBER

`NUMBER(p, s)`

- `NUMERIC` works in Snowflake as an alias for `NUMBER`
- `p` = precision; `s` = scale
- Max `p` and `s` values: 38
 - Exceeding will cause rounding!

Snowflake SQL data types - TIMESTAMP_LTZ

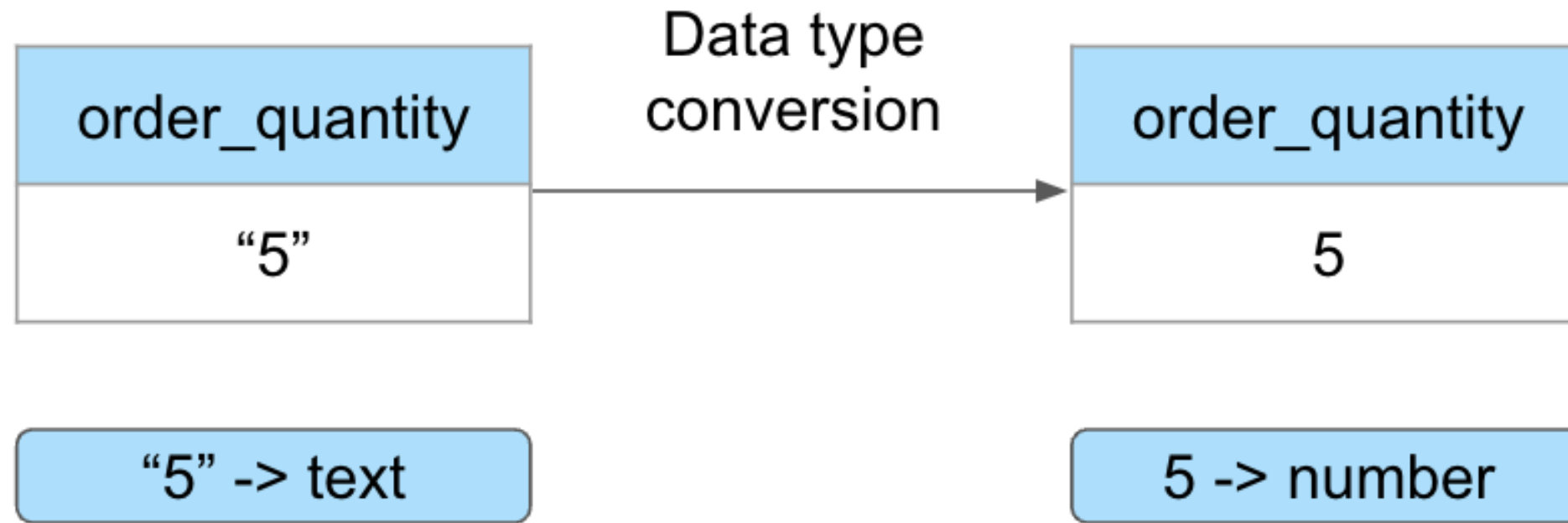
- `TIMESTAMP_LTZ`
 - Combines `DATE` and `TIME` with local time zone
 - Format: `YYYY-MM-DD HH:MI:SS`

```
CREATE TABLE orders (  
  -- Timestamp with local time zone  
  order_timestamp TIMESTAMP_LTZ  
)
```

	ORDER_TIMESTAMP
1	2015-01-01 11:38:36.000 -0800
2	2015-01-01 11:57:40.000 -0800
3	2015-01-01 12:12:28.000 -0800
4	2015-01-01 12:16:31.000 -0800
5	2015-01-01 12:21:30.000 -0800

Data type conversion - What?

- Converting data from one type to another



Data type conversion - Why?

- Improving performance
- Data accuracy and consistency
- Data quality

Data type conversion - How?

1. CAST Syntax:

- `CAST( <source_data/column> AS <target_data_type> )`
- `CAST('80' AS INT)`

2. :: Syntax:

- `<source_data/column>::<target_data_type>`
- `'80'::INT`

CAST

```
SELECT CAST(order_timestamp AS DATE)
       AS order_date
FROM orders
```

CAST results

	ORDER_TIMESTAMP
1	2015-01-01 11:38:36.000 -0800
2	2015-01-01 11:57:40.000 -0800
3	2015-01-01 12:12:28.000 -0800
4	2015-01-01 12:16:31.000 -0800
5	2015-01-01 12:21:30.000 -0800

	ORDER_DATE
1	2015-01-01
2	2015-01-02
3	2015-01-03
4	2015-01-04
5	2015-01-05

Conversion functions

- Examples: `TO_VARCHAR` , `TO_DATE` , etc.

`TO_VARCHAR`

- `TO_VARCHAR( <expr> )`
 - `expr` - numeric, timestamp, etc.
 - Result: `VARCHAR`

Example:

```
SELECT TO_VARCHAR(86)
```

Result:

```
86
```

Checking data types

```
DESC TABLE orders
```

name	type	kind	null?	default	primary key
ORDER_ID	NUMBER(38,0)	COLUMN	N	null	Y
ORDER_DATE	DATE	COLUMN	Y	null	N
ORDER_TIME	TIME(9)	COLUMN	Y	null	N

Let's practice!

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Functions, sorting, and grouping

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String functions - INITCAP

Syntax: `INITCAP( <expr> )`

- Capitalize each word in a string

```
SELECT INITCAP(category) AS capitalized_category
FROM pizza_type
```

<u>A</u> CAPITALIZED_NAMES
The Barbecue Chicken Pizza
The California Chicken Pizza
The Chicken Alfredo Pizza
The Chicken Pesto Pizza

String functions - CONCAT

- Combines the expressions

Syntax:

```
CONCAT( <expr1> [ , <exprN> ... ] )
```

Before Concat:

CATEGORY
Chicken
Classic
Supreme
Veggie

- Combining `category` with ' - Pizza'

```
SELECT CONCAT(category, ' - Pizza')  
  AS pizza_category  
FROM pizza_type
```

After Concat:

PIZZA_CATEGORY
Chicken - Pizza
Classic - Pizza
Supreme - Pizza
Veggie - Pizza

DATE & TIME functions

- `CURRENT_DATE()` or `CURRENT_DATE`
- `CURRENT_TIME()` or `CURRENT_TIME`

```
SELECT CURRENT_DATE
```

```
SELECT CURRENT_TIME
```

CURRENT_DATE
2023-08-15

CURRENT_TIME
13:35:58

EXTRACT

Syntax

- `EXTRACT( <date_or_time_part> FROM <date_or_time_expr> )`
 - `<date_or_time_part>` - `year`, `month`, `day`, `weekday`, etc.

```
SELECT EXTRACT(MONTH FROM order_date) AS order_month,  
       COUNT(*) AS num_orders  
FROM orders  
GROUP BY order_month
```

# ORDER_MONTH	# NUM_ORDERS
1	1845
2	1685
3	1840

SORTING and GROUPING

- SORTING: ORDER BY
- GROUPING: GROUP BY
 - Snowflake: GROUP BY ALL

GROUP BY ALL

- `GROUP BY column1, column2`

SELECT

```
    pizza_type_id,  
    size,  
    AVG(price) AS average_price
```

FROM

```
    pizzas
```

GROUP BY

```
    pizza_type_id, -- explicit columns  
    size
```

ORDER BY

```
    pizza_type_id, average_price DESC
```

- `GROUP BY ALL`

SELECT

```
    pizza_type_id,  
    size,  
    AVG(price) AS average_price
```

FROM

```
    pizzas
```

GROUP BY ALL -- Don't specify columns

ORDER BY

```
    pizza_type_id, average_price DESC
```


Summary

Function/keyword	Use
<code>INITCAP()</code>	Capitalize each word in a string
<code>CONCAT()</code>	Combine multiple strings
<code>CURRENT_DATE</code>	Get the current date
<code>CURRENT_TIME</code>	Get the current time
<code>EXTRACT</code>	Pull a date/time element, e.g., <code>month</code> from a date
<code>ORDER BY</code>	Sort query results
<code>GROUP BY ALL</code>	Group query by all (non-aggregated) columns

Let's practice!

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